

# GLOBAL INDUSTRY CHALLENGE 2026

## TEAM COVER PAGE

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| Challenge Name: Quantum Reservoir Computing for Time-Series Intelligence |           |                           |                 |                                           |
|--------------------------------------------------------------------------|-----------|---------------------------|-----------------|-------------------------------------------|
| Phase #: 3                                                               |           |                           |                 |                                           |
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# Dual-Timescale Quantum Reservoir Computing for Financial Volatility Forecasting

QuantumEdge — Global Industry Challenge 2026, Phase 3

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## Abstract

Financial volatility combines heterogeneous persistence, nonlinear clustering, and abrupt regime changes. We present a dual-timescale quantum reservoir computing (QRC) system for one-day-ahead S&P 500 variance forecasting and calm-versus-turbulent regime detection. Two fixed nine-qubit transverse-field-Ising reservoirs encode fast shocks and slower persistence; only classical ridge and logistic readouts are trained. On a chronological 750-day holdout, the train-calibrated QRC obtains RMSE  $7.532 \times 10^{-5}$  and QLIKE 0.3261, improving over HAR-RV by 5.9% and 21.0%, respectively. Diebold–Mariano tests favor QRC over HAR-RV for squared error ( $p = 0.0028$ ) and QLIKE ( $p = 0.0033$ ). QRC features yield 78.1% two-state regime accuracy versus a 64.7% majority baseline; transition-event F1 reaches 0.707 within a  $\pm 3$ -day tolerance. On authentic Oxford-Man five-minute realized variance, QRC is effectively tied with HAR-RV. Ablation, qubit-count, encoding-density, shot, depolarizing-noise, amplitude-damping, and zero-noise-extrapolation studies characterize robustness without showing monotonic scaling advantage. A targeted seven-qubit IBM Marrakesh run achieves simulator–hardware feature correlation 0.9963 and MAE 0.0304 over 39 Pauli observables. The contribution is a reproducible hardware-ready forecasting challenger, not a claim of quantum computational advantage.

## 1. Problem Framing and Contributions

Volatility forecasts influence derivatives pricing, hedging, market making, risk limits, and execution. The problem is challenging because realized variance is persistent across daily, weekly, and monthly horizons yet reacts abruptly to shocks. HAR-RV explicitly models heterogeneous persistence [1], while recurrent methods learn temporal structure from data. QRC offers a complementary hybrid approach: a fixed quantum dynamical system supplies nonlinear temporal features and a lightweight classical readout is trained [2]. Recent work has demonstrated large-scale gradient-free quantum reservoir learning [3] and applied QRC to realized-volatility forecasting [4]. Feedback-enhanced, few-atom reservoirs with continuous measurement have separately been shown to match or exceed classical reservoir baselines on standard memory and nonlinearity benchmarks [5], and a generalized-synchronization analysis has established conditions under which QRCs remain robust forecasters of chaotic dynamics, including under realistic noise [6].

We target a narrow Track-A question: can a small, hardware-native QRC improve one-day-ahead S&P 500 variance forecasts and identify volatility regimes under reproducible near-term constraints? The study contributes: (i) dual quantum reservoirs matched to fast and slow volatility structure; (ii) an ablation-selected Pauli-only readout deployable on gate-model hardware; (iii) locked comparison against HAR-RV, ESN, LSTM, GARCH, and persistence; (iv) authentic Oxford-Man validation, transition-event analysis, statistical tests, scaling and noise studies; and (v) an auditable qBraid workflow with retained IBM Marrakesh job artifacts.

## 2. Dual-Timescale QRC

### 2.1 Inputs and fixed dynamics

Let  $RV_t$  denote daily variance and  $\ell_t = \log RV_t$ . The five-dimensional input is

$$\mathbf{x}_t = \left[ \ell_t, \bar{\ell}_t^{(5)}, \bar{\ell}_t^{(22)}, \ell_t - \ell_{t-1}, \frac{\text{VIX}_t}{100} - \sqrt{252RV_t} \right], \quad (1)$$

scaled using training data to  $[-\pi/2, \pi/2]$ . A memory vector is appended before repeated angle encoding. For reservoir  $r \in \{s, l\}$ , one repetition applies

$$U_r(\mathbf{x}_t) = \left[ \prod_i R_x^{(i)}(2h_r) \right] \left[ \prod_{i=1}^{n-1} R_{ZZ}^{(i,i+1)}(2J_r) \right] \left[ \prod_i R_y^{(i)}(x_{t,i \bmod d}) \right], \quad (2)$$

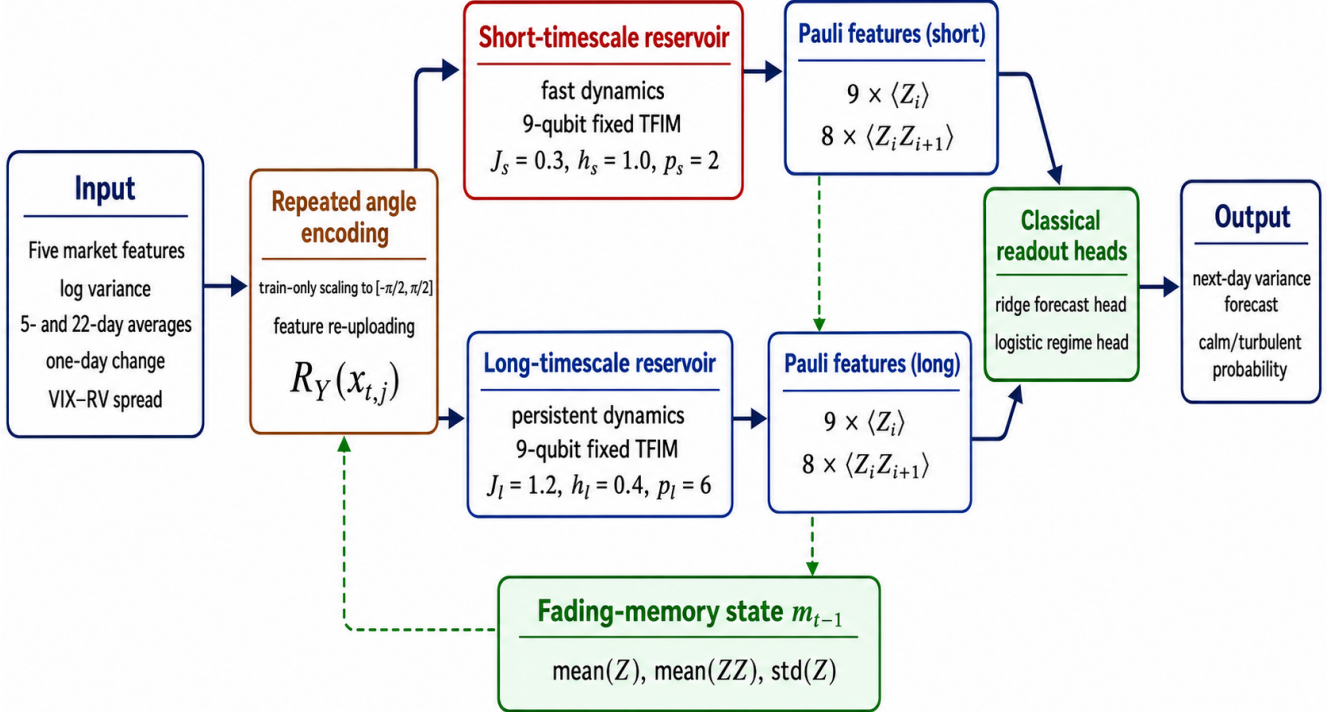
and is repeated  $p_r$  times. Both reservoirs use  $n = 9$  qubits. The short channel uses  $(J_s, h_s, p_s) = (0.3, 1.0, 2)$  and depth 14; the long channel uses  $(J_l, h_l, p_l) = (1.2, 0.4, 6)$  and depth 30. The parameters are fixed rather than variationally optimized.

## 2.2 Hardware-native readout and feedback

Each reservoir returns  $n$  single-qubit features  $\langle Z_i \rangle$  and  $n - 1$  nearest-neighbor features  $\langle Z_i Z_{i+1} \rangle$ . Concatenating two reservoirs gives 34 features. A three-dimensional summary  $\mathbf{s}_t = [\text{mean}(Z), \text{mean}(ZZ), \text{std}(Z)]$  updates fading memory as

$$\mathbf{m}_t = (1 - \gamma)\mathbf{m}_{t-1} + \gamma\mathbf{s}_t, \quad \gamma = 0.5. \quad (3)$$

A standardized ridge head predicts next-day log variance. RidgeCV selects its regularization parameter; the executed run chose  $\alpha = 0.3162$ . An affine correction is fit only on training predictions in variance space and then frozen for test inference. A separate logistic head uses the same QRC features for calm/turbulent classification, where the threshold is the training-sample median variance.



**Figure 1:** QuantumEdge architecture. Quantum circuits are fixed; learning occurs only in the classical heads and optional train-only calibration.

## 3. Data and Experimental Protocol

The committed market file contains 3,773 trading days from 5 January 2010 through 31 December 2024. S&P 500 OHLC prices produce Garman–Klass variance [7]; VIX and NFCI originate from FRED, with NFCI retained for provenance but excluded from the locked five-feature vector. After the 22-day window, 3,750 samples remain. The first 3,000 are training observations and the final 750 are an untouched chronological test set. No random shuffling is used. Authentic transfer is evaluated using Oxford-Man S&P 500 five-minute realized volatility, squared to variance when required, with 519 held-out observations [8].

Classical comparators are HAR-RV, a 200-node ESN [9], an early-stopped 64-hidden-unit LSTM [10], GARCH(1,1) [11], and persistence. Primary losses are

$$\text{RMSE} = \sqrt{T^{-1} \sum_t (RV_t - \widehat{RV}_t)^2}, \quad \text{QLIKE} = T^{-1} \sum_t \left( \frac{RV_t}{\widehat{RV}_t} - \log \frac{RV_t}{\widehat{RV}_t} - 1 \right), \quad (4)$$

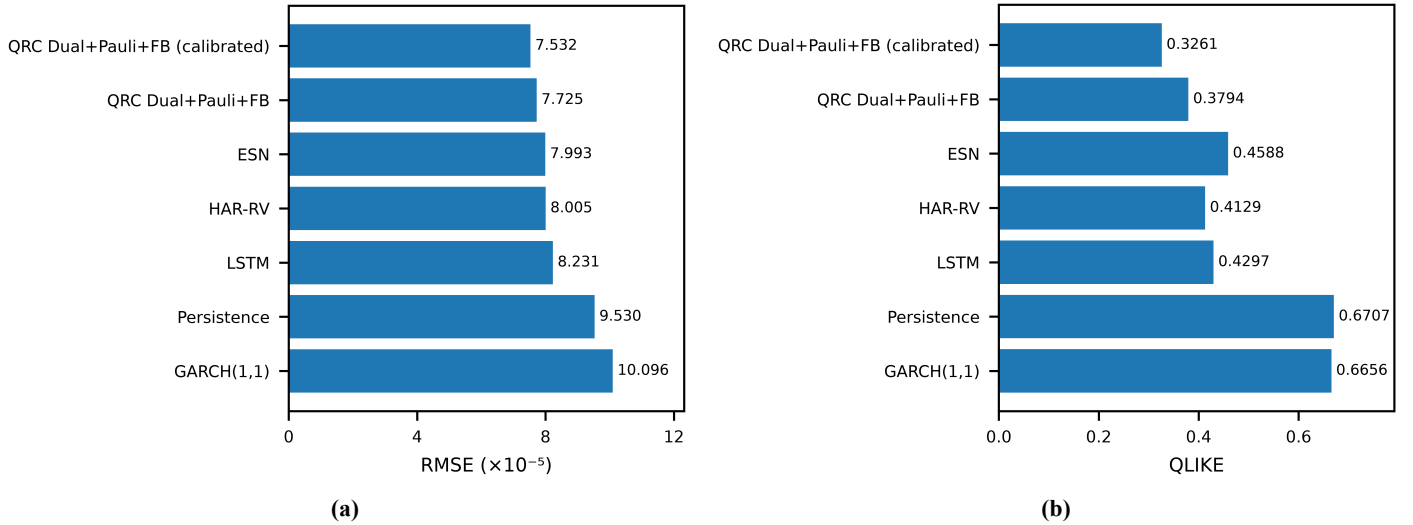
where QLIKE is robust to noisy volatility proxies [12]. We additionally report Mincer–Zarnowitz calibration [13], two- and three-state regime accuracy, transition-event metrics, a secondary volatility-targeting Sharpe statistic, Diebold–Mariano tests [14], and moving-block-bootstrap intervals. All preprocessing, calibration, and model selection use training data only.

#### 4. Forecasting and Regime Results

**Table 1:** Locked 750-day one-step test. Lower RMSE and QLIKE are better; an MZ slope near one is preferred.

| Model                         | RMSE ( $\times 10^{-5}$ ) | QLIKE         | MZ slope |
|-------------------------------|---------------------------|---------------|----------|
| QRC dual+Pauli+FB, calibrated | <b>7.532</b>              | <b>0.3261</b> | 1.465    |
| QRC dual+Pauli+FB, raw        | 7.725                     | 0.3794        | 1.298    |
| ESN                           | 7.993                     | 0.4588        | 1.101    |
| HAR-RV                        | 8.005                     | 0.4129        | 1.262    |
| LSTM                          | 8.231                     | 0.4297        | 1.630    |
| Persistence                   | 9.530                     | 0.6707        | 0.453    |
| GARCH(1,1)                    | 10.096                    | 0.6656        | -3.098   |

The calibrated QRC leads both primary losses (Table 1), reducing RMSE by 5.9% and QLIKE by 21.0% relative to HAR-RV. The raw QRC also leads the uncalibrated methods, so the ranking is not created only by affine correction. Against HAR-RV, Diebold–Mariano statistics are  $-2.994$  for squared error ( $p = 0.0028$ ) and  $-2.936$  for QLIKE ( $p = 0.0033$ ); negative loss differentials favor QRC. Moving-block-bootstrap intervals for QRC minus HAR-RV exclude zero for RMSE ( $[-6.43, -2.05] \times 10^{-6}$ ) and QLIKE ( $[-0.111, -0.031]$ ). These results support a sample-specific forecasting lift, not a universal or computational quantum advantage. Residual bias remains: the QRC MZ slopes exceed one, and the LSTM has the largest secondary Sharpe (0.499 versus 0.475 for calibrated QRC).



**Figure 2:** Forecasting performance on the locked chronological test set: (a) root-mean-square error and (b) quasi-likelihood loss. Lower values indicate better performance.

The dedicated binary head attains 78.1% calm/turbulent accuracy versus 64.7% for a majority rule. There are 212 true and 64 predicted transitions. Within  $\pm 1$  day, event precision/recall/F1 are 0.922/0.377/0.536; within  $\pm 3$  days they are 1.000/0.547/0.707. Accuracy restricted to transition windows is 59.6% and 68.8% for the two tolerances, compared with majority-window accuracies 51.4% and 53.1%. Thus the features carry useful regime information, although transition recall remains moderate.

On authentic Oxford-Man variance, QRC and HAR-RV are effectively tied: QRC obtains  $RMSE\ 5.594 \times 10^{-5}$  and  $QLIKE\ 0.2425$ , versus  $5.599 \times 10^{-5}$  and 0.2351 for HAR-RV; both beat persistence. We therefore claim transfer competitiveness, not statistical superiority on authentic intraday variance.

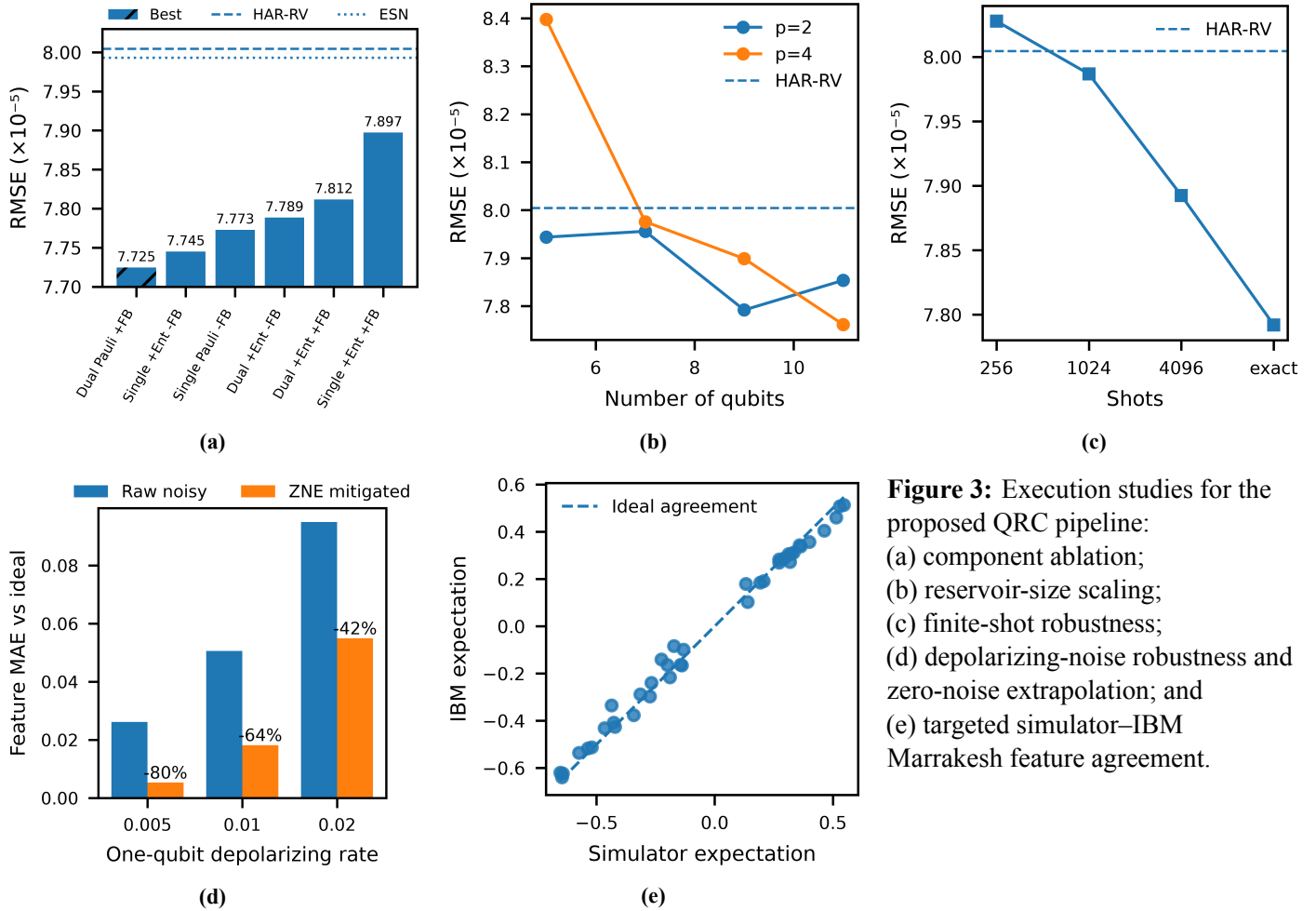
#### 5. Execution Studies and IBM Marrakesh Validation

Ablation identifies the raw dual Pauli+feedback model as the best uncalibrated variant ( $RMSE\ 7.725 \times 10^{-5}$ ,  $QLIKE\ 0.3794$ ). Adding simulator-only entanglement-spectrum features does not help; the best such variant reaches  $RMSE\ 7.745 \times 10^{-5}$ . The dual reservoir only marginally improves over single-reservoir variants, and feedback is approximately neutral across matched settings. This null result for feedback contrasts with feedback-driven gains reported in continuous-measurement cavity-QED reservoirs [5], suggesting that the benefit of measurement feedback is architecture- and task-

dependent rather than universal in QRC design. These negative results justify the simpler hardware-native model.

Across 5, 7, 9, and 11 qubits with two or four repetitions, RMSE remains approximately  $[7.76, 8.40] \times 10^{-5}$  with no monotonic improvement. Encoding 3, 6, and 9 qubits gives RMSE  $8.019, 7.948$ , and  $7.792 \times 10^{-5}$ . Finite-shot RMSE improves from  $8.028 \times 10^{-5}$  at 256 shots to  $7.893 \times 10^{-5}$  at 4,096 shots, approaching the exact value  $7.792 \times 10^{-5}$ .

For depolarizing error rates  $(p_{1q}, p_{2q}) = (0.005, 0.010)$ ,  $(0.010, 0.020)$ , and  $(0.020, 0.040)$ , raw feature MAE is 0.0261, 0.0506, and 0.0950. Richardson-style zero-noise extrapolation [15] reduces MAE to 0.0053, 0.0182, and 0.0549, recovering approximately 80%, 64%, and 42% of induced error. With a fixed 0.005/0.010 depolarizing background, amplitude damping  $\gamma = 0.001$ –0.020 increases feature MAE from 0.0271 to 0.0470. These noise-robustness trends are consistent with generalized-synchronization analyses showing that QRCs can remain reliable forecasters of chaotic dynamics under realistic noise provided the reservoir stays within its stable synchronization regime [6].



**Figure 3:** Execution studies for the proposed QRC pipeline: (a) component ablation; (b) reservoir-size scaling; (c) finite-shot robustness; (d) depolarizing-noise robustness and zero-noise extrapolation; and (e) targeted simulator-IBM Marrakesh feature agreement.

The final auditable QPU execution used the seven-qubit IBM Marrakesh backend. Three representative validation windows were transpiled at optimization level 3 to depths 71, 70, and 70 and executed with 4,096 shots per circuit. Across 39  $Z/Z$  observable pairs, the simulator-hardware Pearson correlation is 0.9963 and the feature MAE is 0.0304; the recorded wall time is 73.65 seconds. The retained job identifiers, per-window ledger, JSON summary, and observable-level CSV all correspond to IBM Marrakesh. This targeted QPU run validates transfer fidelity of the hardware-native features consumed by the classical readout; it is not a full 750-day QPU forecast or a runtime-advantage demonstration.

The required canonical-MNIST check [16] uses the same fixed-Ising/Pauli concept on a balanced 100-train/50-test subset. Accuracy is 32%, 60%, and 58% for 5, 10, and 15 qubits, versus 80% for a pooled-pixel logistic control. We report this negative result as a cross-domain expressivity check, not evidence of general image-classification advantage.

## 6. Stakeholder Value, Limitations, and Reproducibility

For a trading or risk desk, the near-term value is model diversification: QRC can operate as a challenger beside HAR-RV and LSTM, with forecast loss, bias, transition recall, stability, and cost thresholds imposed before any capital decision. The fixed reservoir eliminates gradient-based quantum training, but production adoption would still require data-lineage

controls, drift and stress tests, explainability of the classical head, hardware-cost benchmarking, and fallback to approved classical models.

The evidence is deliberately bounded. The headline improvement is modest and sample-specific; MZ tests reveal residual bias; Oxford-Man performance is a tie; the tested qubit range is flat; entanglement features do not help; MNIST underperforms its classical control; and hardware validation covers only three representative windows. The study is limited to the S&P 500 and one-day horizon.

The qBraid-ready package fixes seed 7 and includes committed public-data mirrors and hashes, cached or rebuildable Pauli features, clean and executed notebooks, exact dependency files, an agent-executable qBraid Skill, forecast and significance CSVs, noise/transition/MNIST artifacts, IBM Marrakesh job evidence, and a package verifier. Judges can run a quick smoke test or the full deterministic workflow without QPU credentials; retained QPU results are loaded by default. Before submission, the repository verifier must report zero failures and the official unmodified GIC cover page must be prepended.

**Conclusion.** QuantumEdge demonstrates that a small dual-timescale QRC can provide a reproducible forecasting lift on the locked variance proxy, useful regime information, competitive authentic-RV performance, robustness under practical shot/noise budgets, and high targeted IBM Marrakesh feature fidelity. The supported position is a hardware-ready quantum forecasting challenger with clear limits, not proven quantum advantage.

**Generative-AI disclosure.** Generative AI assisted with code drafting, debugging, documentation structure, language editing, and reproducibility materials. The team selected the research question, architecture, data, baselines, experimental protocol, and hardware plan; executed and verified all experiments; interpreted all findings; and accepts responsibility for every claim. No AI-generated number is reported as experimental evidence.

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